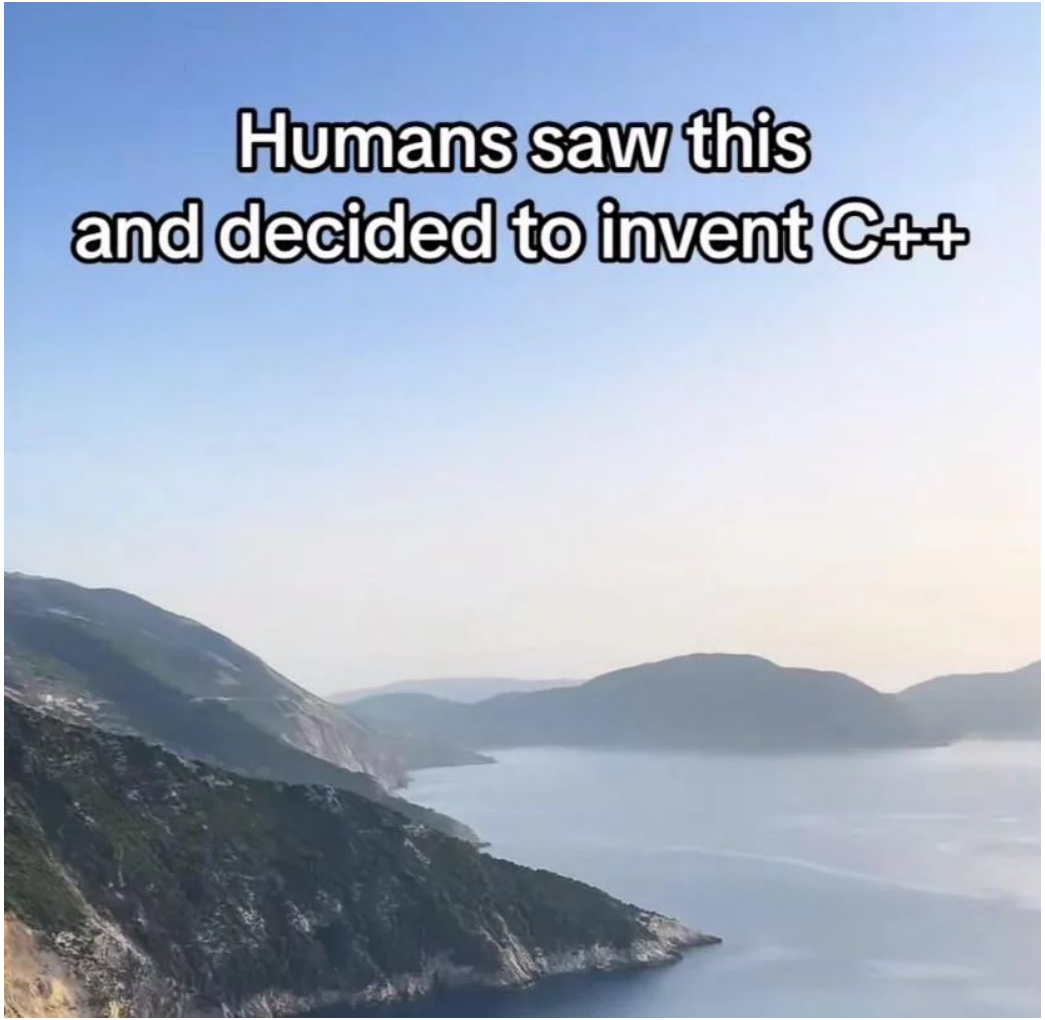
CS 103000
Prof. Madeline Blount

Week 10



Dall-E 2: cats learning C++ in the forest on ‘90’s technology


Humans saw this
and decided to invent C++

strong projects: A/A-

- most robust methods for surfacing key (frequency analysis)
- creative, unique encryption methods
- clear, well-marked comments, can follow thought process
- considered front-end user: menu experience, personality
- strong understanding during oral presentation
- error handling
- CITATIONS! robust

solid projects: B/B+

- completed part 1 + 2
- loaded in both string inputs via stream
- basic but functional riff on an LCG for PRNG
- general understanding during oral presentation
- citations but lacking full info
- some style issues: indentation, unclear names, etc.

needs improvement projects:

- incomplete or partial part 1 or 2
- unnecessary vars, libraries, vector of alphabets, etc.
- unclear understanding during oral presentation
- poor citation or overuse of external/AI tools

🌍 CLASSES + OBJECTS = NEW PARADIGM

- Procedural programming
- Functional programming
- Object-oriented programming (OOP)
  - *More* abstraction
  - *More* control
  - ENCAPSULATION



🌍 **OBJECT ORIENTED = NEW PARADIGM, classes**

- Bjarne Stroustrup made C++ with classes in mind
- CLASS = **make your own data type**
- WHEN make a class?
  - When some data type will repeat
  - When your data will have multiple attributes
  - “Model” something in the world

codecademy









Lightbulb
isOn *(boolean)*
watts *(number)*
turnOn() *(func)*

Microphone
color *(string)*
isAnalog *(boolean)*
record() *(func)*

Animal
color *(string)*
age *(number)*
speak() *(func)*

Date
month *(number)*
year *(number)*
toString() *(func)*

CLASS vs. OBJECT

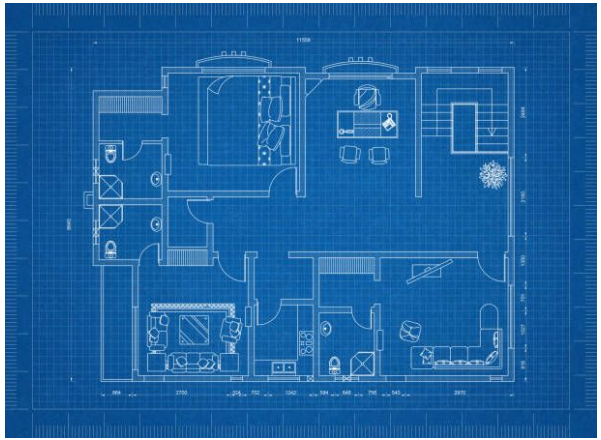
- **Class is the blueprint; object is an INSTANCE of the class**

CLASS

OBJECT

blueprint

building





CLASS vs. OBJECT

- **Class is the blueprint; object is an INSTANCE of the class**

CLASS

cookie cutter



OBJECT

cookie



CLASS vs. OBJECT

- **Class is the blueprint; object is an INSTANCE of the class**

CLASS

OBJECT

species



Actual birds in the world



MAKING A CLASS

- Class definition
- Attributes, or member data (what are the THINGS that define the objects in this class? Size, name, color, rate, cost, etc.)
- Methods, or member functions (what are the things you can DO with the objects in this class)

FILE STRUCTURE

imagine class called Movie…



main.cpp

#include



Movie.h

Class encapsulation, (attributes, member data, declarations)

#include



Movie.cpp

Class methods (function definitions)

compile: `g++ main.cpp Movie.cpp`



CONSTRUCTOR

- object initialization function, runs automatically
- declared in **public** encapsulation
- Funny definitions syntax … **DOES NOT HAVE RETURN TYPE!**

  example: class named Neighbor

```
Neighbor::Neighbor() {

}
```



STRENGTH in COMMUNITY

Crown Heights Mutual Aid

Join a network of neighbors providing for one another

CROWN HEIGHTS MUTUAL AID is a network of neighbors supporting one another and the most vulnerable in our community in the face of this pandemic and the crises we face from climate change, state violence, and inequality. We all have something to offer and we all have something we need.

Join us at crownheightsmutualaid.com/slack or on Facebook at Crown Heights Mutual Aid

What we had:

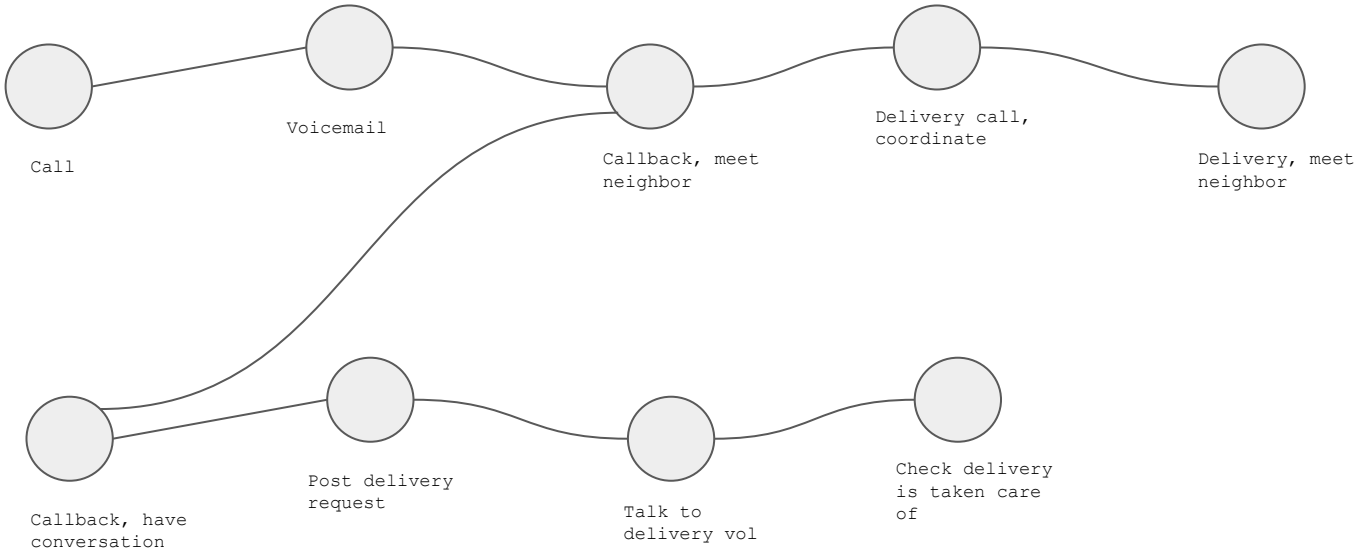
- **Need** list (growing) - turning to new places, everything shut
- **Volunteer base** (growing) - time, interest: delivery, phone calls, donations
- **Technical** savvy, for some (Slack, Facebook, Twitter, etc. - so much time on these platforms)
- **Small team** of coders (3-20), larger group using system
- **Security minded**

What we needed:

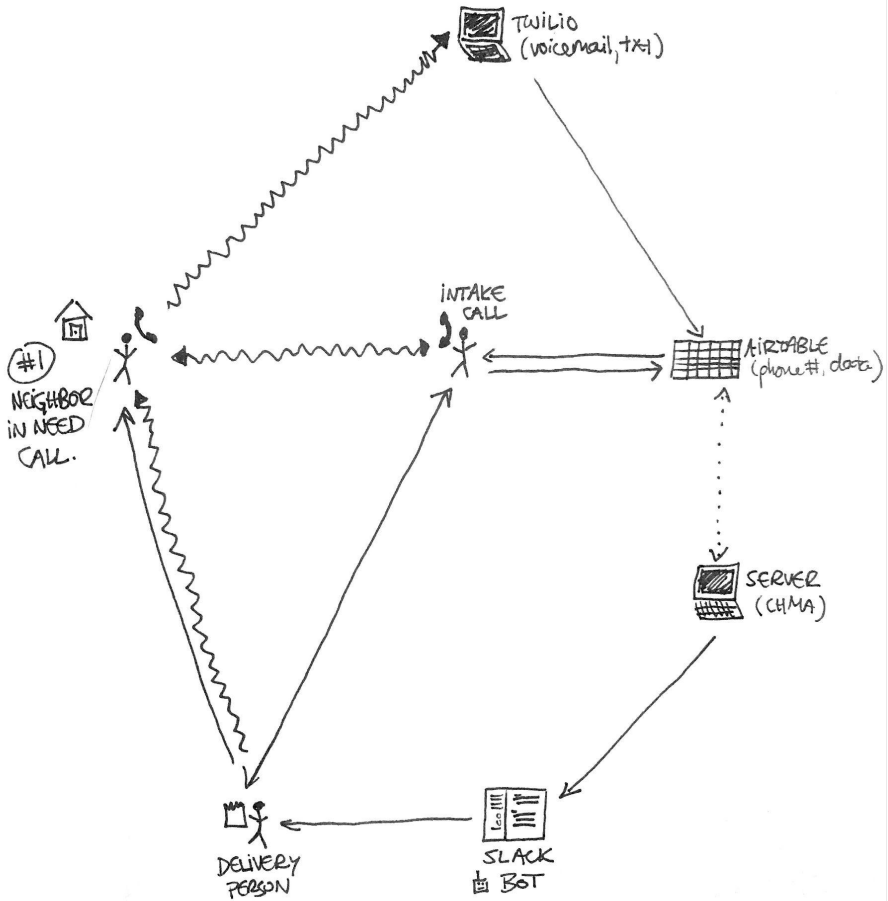
- Way to **connect** local neighbors in need with local neighbors able to give

Systems/JOURNEYS: user/data

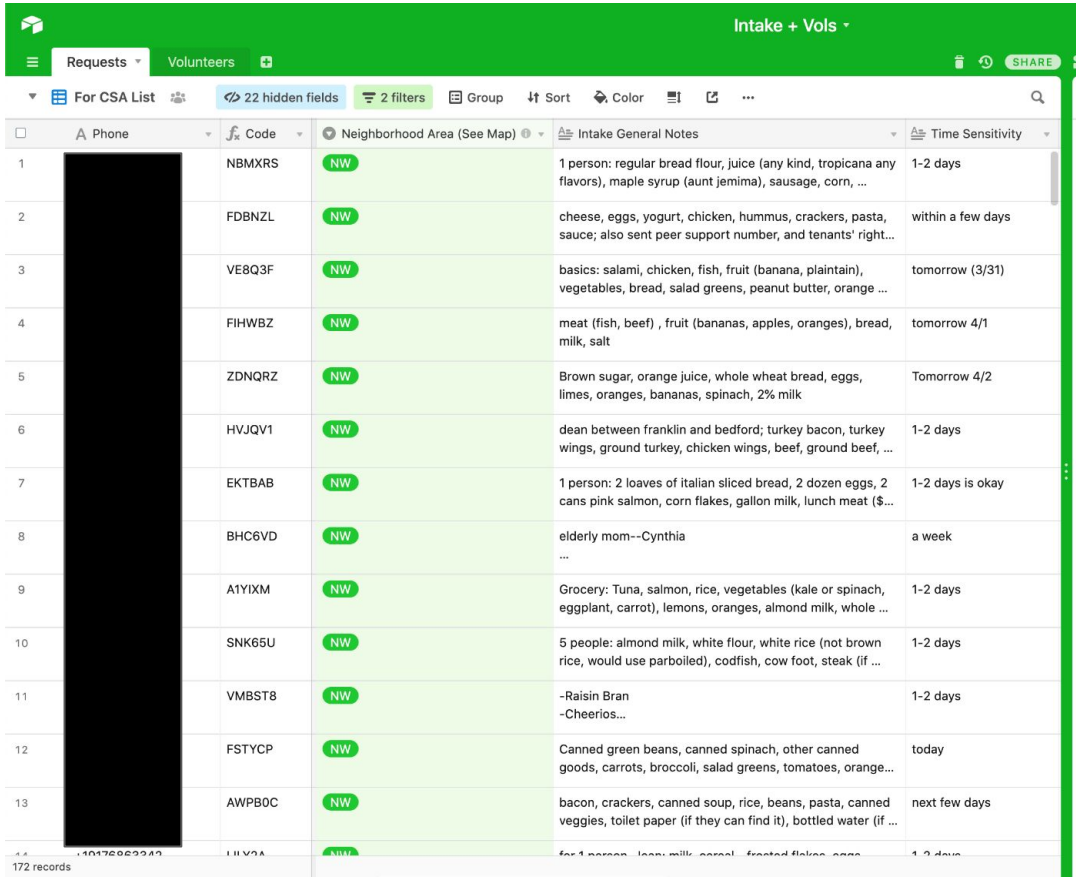
[Neighbor in Need]



[Intake Volunteer]


TWILIO
(voicemail, txt)
INTAKE
CALL
AIRTABLE
(phone#, data)
#1
NEIGHBOR
IN NEED
CALL.
SERVER
(CHMA)
DELIVERY
PERSON
SLACK
BOT

Step 2: database, intake

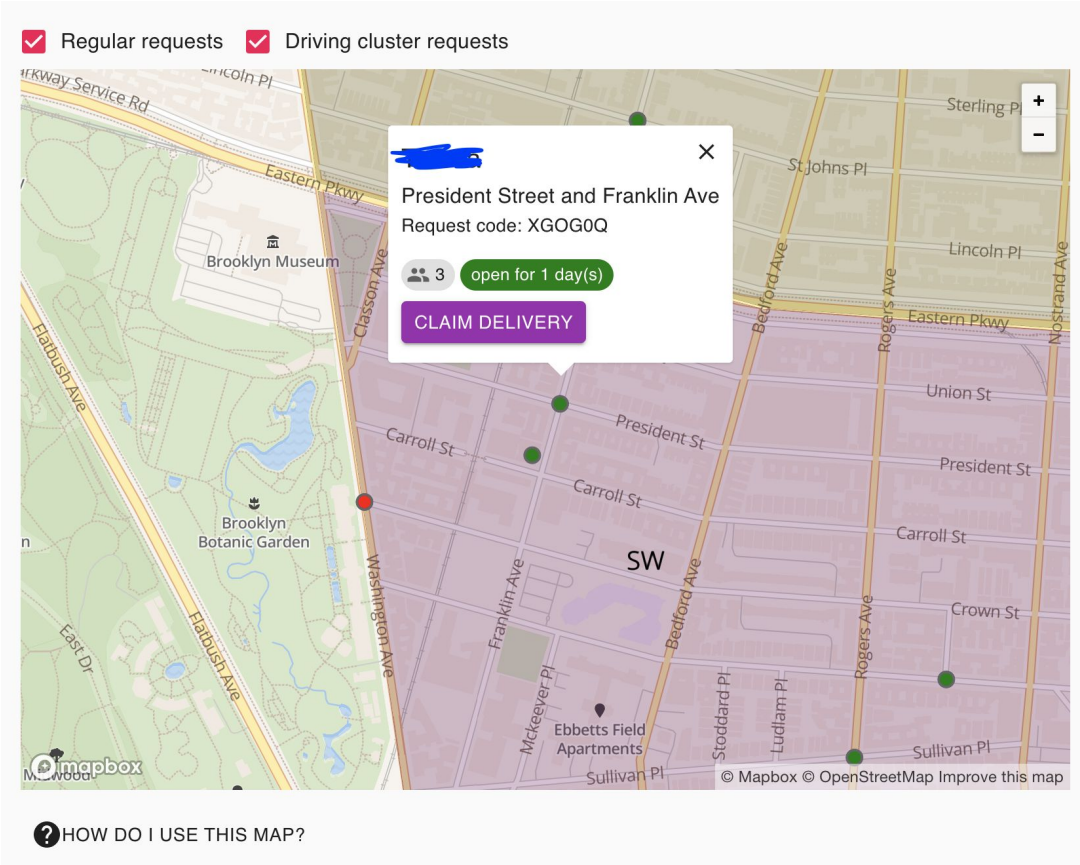


	Phone	Code	Neighborhood Area (See Map)	Intake General Notes	Time Sensitivity
1	[redacted]	NBMXRS	NW	1 person: regular bread flour, juice (any kind, tropicana any flavors), maple syrup (aunt jemima), sausage, corn, ...	1-2 days
2		FDBNZL	NW	cheese, eggs, yogurt, chicken, hummus, crackers, pasta, sauce; also sent peer support number, and tenants' right...	within a few days
3		VE8Q3F	NW	basics: salami, chicken, fish, fruit (banana, plaintain), vegetables, bread, salad greens, peanut butter, orange ...	tomorrow (3/31)
4		FIHWBZ	NW	meat (fish, beef) , fruit (bananas, apples, oranges), bread, milk, salt	tomorrow 4/1
5		ZDNQRZ	NW	Brown sugar, orange juice, whole wheat bread, eggs, limes, oranges, bananas, spinach, 2% milk	Tomorrow 4/2
6		HVJQV1	NW	dean between franklin and bedford; turkey bacon, turkey wings, ground turkey, chicken wings, beef, ground beef, ...	1-2 days
7		EKTBAB	NW	1 person: 2 loaves of italian sliced bread, 2 dozen eggs, 2 cans pink salmon, corn flakes, gallon milk, lunch meat ($...	1-2 days is okay
8		BHC6VD	NW	elderly mom--Cynthia ...	a week
9		A1YIXM	NW	Grocery: Tuna, salmon, rice, vegetables (kale or spinach, eggplant, carrot), lemons, oranges, almond milk, whole ...	1-2 days
10		SNK65U	NW	5 people: almond milk, white flour, white rice (not brown rice, would use parboiled), codfish, cow foot, steak (if ...	1-2 days
11		VMBST8	NW	-Raisin Bran -Cheerios...	1-2 days
12		FSTYCP	NW	Canned green beans, canned spinach, other canned goods, carrots, broccoli, salad greens, tomatoes, orange...	today
13		AWPB0C	NW	bacon, crackers, canned soup, rice, beans, pasta, canned veggies, toilet paper (if they can find it), bottled water (if ...	next few days

172 records

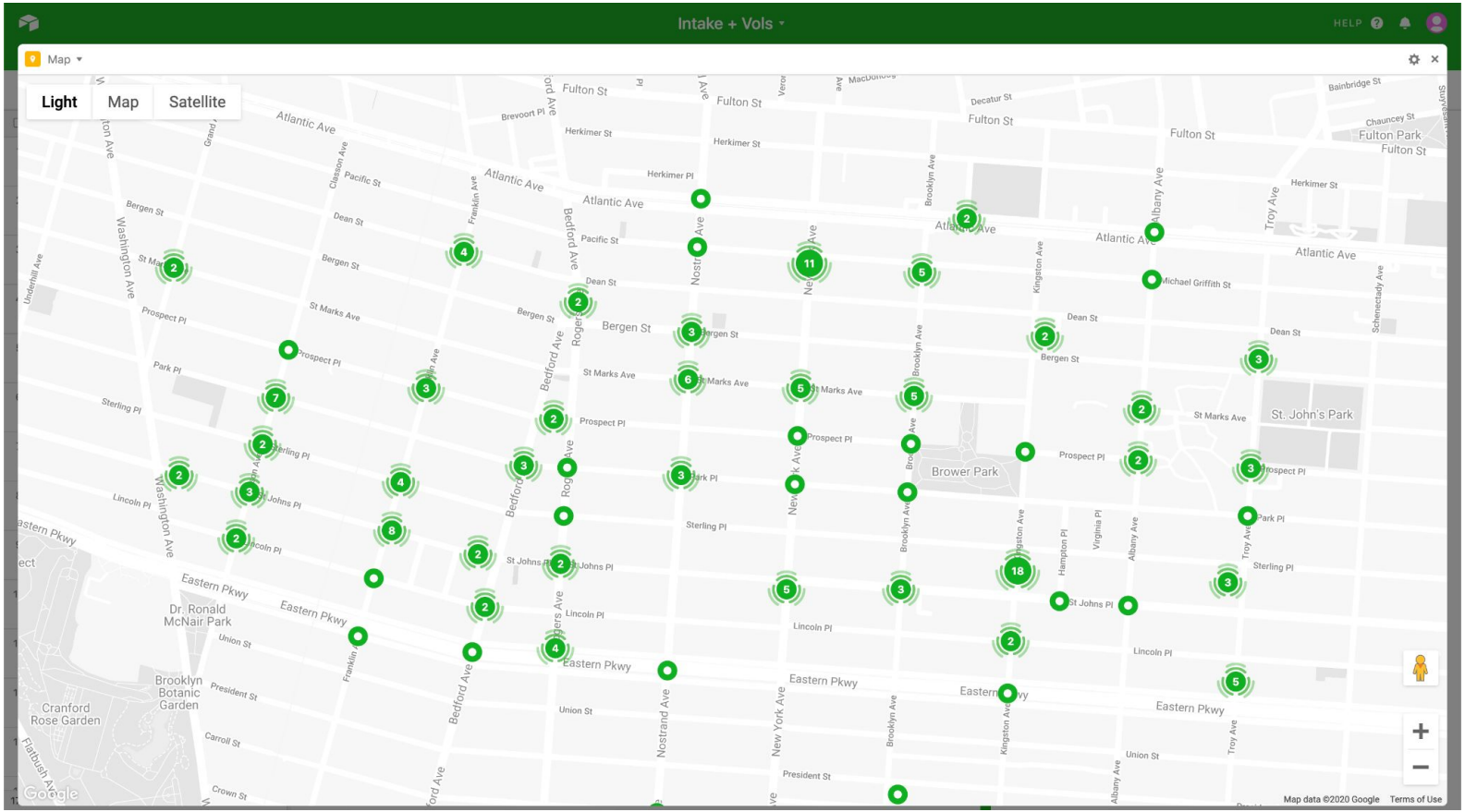
- "Intake" volunteer, uses the **relational** database (Airtable) like a call sheet; unique code generated
- First connection made: neighbor in need to neighbor on the phone (shifts, parties)
- Fills out fields that are empty from phone conversation
- **DATA SECURITY:** vulnerable; but only keeps first names, and cross street
- Volume …

Version 2.0 delivery system: web map

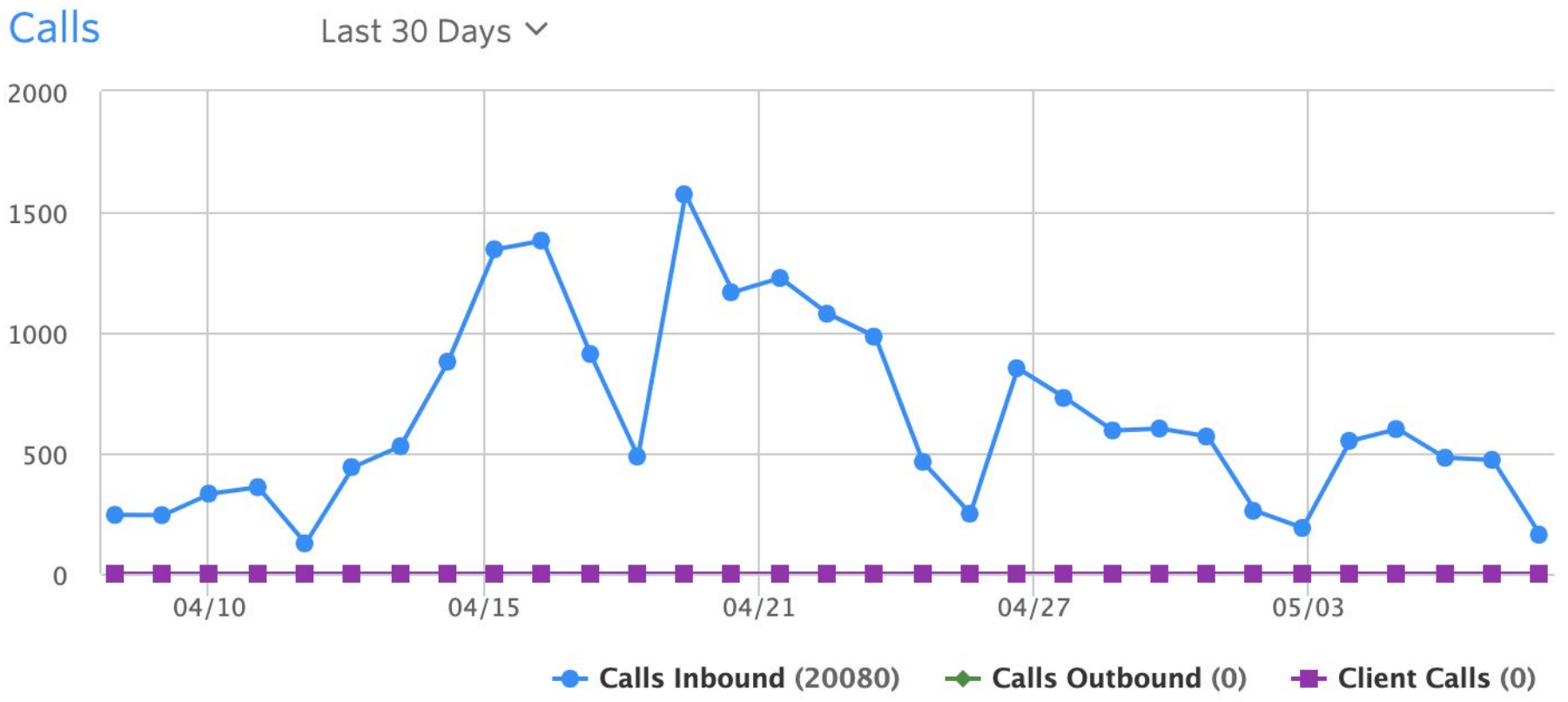


- When intake volunteer changed status column in Airtable to "Needs Delivery" call sent to endpoint on Node.js server
- Used Google Maps API to geolocate cross streets, place them on web map (front-end: Mapbox)

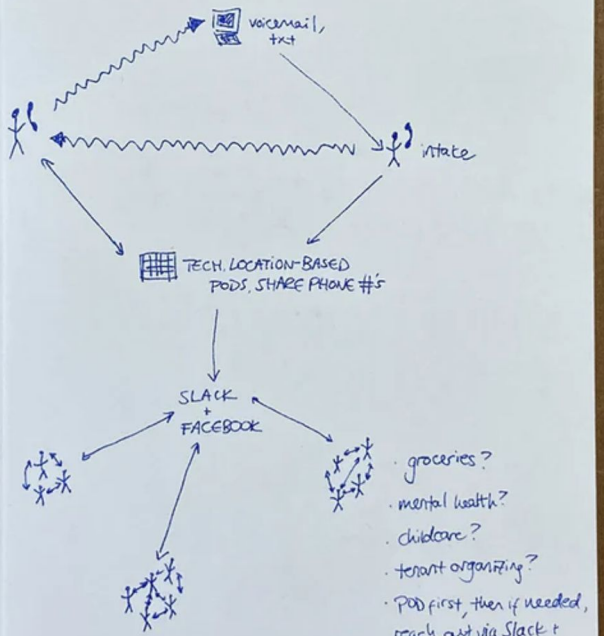
Extensions & Projects:



SCALE SCALE SCALE SCALE SCALE SCALE



Rehearsing Solidarity
Learning from Mutual Aid




WE THE PEOPLE
MUST HELP
EACH OTHER!